

Group 2: Ralchukwu Mogbogu & Yiye Sun

Preprocess and Visualize

ANLY 540

9/16/2024

```
!pip install textacy
!pip install plotly
```

Show hidden output

```
import pandas as pd
import os
import string
from string import digits
import re
import unicodedata
```

```
#Import the Jobs.csv file
from google.colab import files
uploaded = files.upload()
```

Choose File

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Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to

```
drugsdf = pd.read_csv('DrugReview.csv', encoding = 'utf8')
```

```
drugsdf.head()
```

Show

		drugName	condition	review	rating	date	usefulCount
0	163740	Mirtazapine	Depression	"I've tried a few antidepressants over th...	10.0	February 28, 2012	22
1	206473	Mesalamine	Crohn's Disease, Maintenance	"My son has Crohn's disease and has done ...	8.0	May 17, 2009	17
2	159672	Bactrim	Urinary Tract Infection	"Quick reduction of symptoms"	9.0	September 29, 2017	3

```
##subsetting because of large data; reducing to 2000 to avoid high memory usage
drugsdf = drugsdf.iloc[:2000]
```

```
import textacy.preprocessing as tprep
def normalize(review):
    review = tprep.normalize.hyphenated_words(review)
    review = tprep.normalize.quotation_marks(review)
    review = tprep.normalize.unicode(review)
    review = tprep.remove.accents(review)
    review = tprep.replace.phone_numbers(review)
    review = re.sub(r'http\S+', '', review) ##changed to this because tprep.replace did not work
    review = tprep.replace.emails(review)
    review = tprep.replace.user_handles(review)
    review = tprep.replace.emojis(review)
    review = review.lower()
    review = unicodedata.normalize('NFKD', review).encode('ascii', 'ignore').decode('utf-8', 'ignore')
    review = re.sub(r'<[^>]*>', ' ', review)
    return review
```

1. Preprocess the text in your corpus.
1. Normalize encodings

2. Lowercase text

3. Remove numbers and punctuation

4. Remove stopwords
2. Summarize all the metadata variables.

3. Create a word cloud of the text.
4. What NLP techniques you could apply to the dataset and what could you learn from them? (**at least 1 full paragraph**)
5. Compile in a notebook and submit html or pdf file.

```
# 1. Normalize encodings
drugsdf['review'] = drugsdf['review'].apply(lambda x: unicodedata.normalize('NFKD', x).encode('ascii', 'ignore').decode('utf-8', 'ignore'))
# 2. Lowercase
drugsdf['review'] = drugsdf['review'].str.lower()
# 3. Remove punctuations and numbers
drugsdf['review'] = drugsdf['review'].str.replace(r'^[\w\s]', '', regex = True)
drugsdf['review'].head(5)

drugsdf['review'] = drugsdf['review'].str.replace('\d+', '')
drugsdf['review'].head(5)

drugsdf['review'] = drugsdf['review'].str.replace(f"[{string.punctuation}0-9]", ' ', regex=True)
drugsdf['review'].head(5)

print(drugsdf[0:5])
```

```
↵
```

	drugName	condition	review	rating
0	163740 Mirtazapine	Depression	i ve tried a few antidepressants over the ye...	10.0
1	206473 Mesalamine	Crohn's Disease, Maintenance	my son has crohn s disease and has done very...	8.0
2	159672 Bactrim	Urinary Tract Infection	quick reduction of symptoms	9.0
3	39293 Contrave	Weight Loss	contrave combines drugs that were used for alc...	9.0
4	97768 Cyclofem 1 / 35	Birth Control	i have been on this birth control for one cycl...	9.0

	date	usefulCount
0	February 28, 2012	22
1	May 17, 2009	17
2	September 29, 2017	3
3	March 5, 2017	35
4	October 22, 2015	4

```
##stopwords
import nltk
nltk.download('stopwords')
from nltk.corpus import stopwords
stopwords = set(stopwords.words("english"))
```

```
↵ [nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
```

```
def remove_stopwords(text):
# return ' '.join([word for word in text.split() if word not in stopwords])
return ' '.join(word for word in text.split() if word not in stopwords)
```

```
# 4.Remove stopwords
drugsdf['review'] = drugsdf['review'].apply(remove_stopwords)
drugsdf.head(5)
```

```
↵
```

	drugName	condition	review	rating	date	usefulCount
0	163740 Mirtazapine	Depression	tried antidepressants years citalopram fluoxet...	10.0	February 28, 2012	22
1	206473 Mesalamine	Crohn's Disease, Maintenance	son crohn disease done well asacol complaints ...	8.0	May 17, 2009	17
2	159672 Bactrim	Urinary Tract Infection	quick reduction symptoms	9.0	September 29, 2017	3
3	39293 Contrave	Weight Loss	contrave combines drugs used alcohol smoking o...	9.0	March 5, 2017	35

```
# Summarize all the metadata variables.
##summary
#Date
summary_date = drugsdf['date'].value_counts(ascending = False)
print(summary_date)
```

```

date
July 23, 2015      7
January 18, 2017   6
December 16, 2016  5
April 29, 2016    5
June 27, 2016     5
..
March 5, 2008      1
October 31, 2009   1
August 27, 2009    1
March 8, 2015      1
February 10, 2012  1
Name: count, Length: 1414, dtype: int64

```

```

summary_dnum = drugsdf['date'].astype(str).describe()
print(summary_dnum)

```

```

count      2000
unique     1414
top      July 23, 2015
freq              7
Name: date, dtype: object

```

```

#rating
sum_r = drugsdf['rating'].value_counts(ascending = False)
print(sum_r)
sum_rnum = drugsdf['rating'].astype(str).describe()
print(sum_rnum)

```

```

rating
10.0    621
9.0     359
1.0     264
8.0     226
7.0     113
5.0      98
2.0      91
3.0      91
6.0      77
4.0      60
Name: count, dtype: int64
count      2000
unique       10
top       10.0
freq       621
Name: rating, dtype: object

```

```

#usefulCount
sum_uc = drugsdf['usefulCount'].value_counts(ascending = False)
print(sum_uc)
sum_ucnum = drugsdf['usefulCount'].astype(str).describe()
print(sum_ucnum)

```

```

usefulCount
4      83
3      82
2      78
5      78
8      77
..
118     1
122     1
384     1
130     1
120     1
Name: count, Length: 161, dtype: int64
count      2000
unique      161
top         4
freq       83
Name: usefulCount, dtype: object

```

```

#condition
sum_c = drugsdf['condition'].value_counts(ascending = False)
print(sum_c)
sum_cnum = drugsdf['condition'].astype(str).describe()
print(sum_cnum)

```

```

condition
Birth Control      369
Depression         128
Pain               76
Acne               67
Anxiety            60
...
me                1
Ulcerative Colitis, Active 1
Human Papillomavirus Prophylaxis 1
Postoperative Pain 1
Secondary Hyperparathyroidism 1
Name: count, Length: 271, dtype: int64
count            2000
unique           272
top      Birth Control
freq            369
Name: condition, dtype: object

```

```

#drugName
sum_dn = drugsdf['drugName'].value_counts(ascending = False)
print(sum_dn)
sum_dnum = drugsdf['drugName'].astype(str).describe()
print(sum_dnum)

```

```

drugName
Levonorgestrel      51
Etonogestrel        49
Ethinyl estradiol / norethindrone 33
Ethinyl estradiol / levonorgestrel 24
Ethinyl estradiol / norgestimate  22
..
Ziana               1
Brovana             1
Horizant            1
Monistat 7          1
Asacol              1
Name: count, Length: 713, dtype: int64
count            2000
unique           713
top      Levonorgestrel
freq            51
Name: drugName, dtype: object

```

```
# Create a word cloud of the text.
```

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt
```

```
all_tokens = [token for sublist in drugsdf['review'] for token in sublist]
all_tokens_text = ''.join(all_tokens)
```

```
all_tokens_text
wordcloud = WordCloud(width=800, height=400, background_color='white', max_words=100).generate(all_tokens_text)
```

```
# Display the word cloud
```

```
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Word Cloud of Drug Review')
plt.show()
```



Semantic Analysis and Topic Modelling are good NLP techniques that could be applied to the data. Especially seeing as the wordclouds bring up common topics, mentions particular phrases, and hints on sentiments being expressed from the reviews.

We could also do some comparative analysis by comparing the topics and sentiments related to the different conditions. We could learn from this form of analysis, a good way to examine patient feedback (or feedback in general) regarding various drugs, while also peeking into the patterns that revolve around patient reviews. Furthermore, we could also analyze the impact of the "usefulCount" to see which reviews to focus on based on how they were found helpful by other patients; this could provide important information in improving patients' usage outcomes for various conditions being addressed.

https://colab.research.google.com/drive/1m69GPRoTXHrzlaRFYKxLTXeEPqVHdaxp#scrollTo=njy_rKd6kZSq&printMode=true